Problem Set 4

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Likelihood Function

The model to be estimated is:

$\ln(w_{i,t}) = \alpha + \beta_1 Educ_{i,t} + \beta_2 Age_{i,t} + \beta_3 Age^2_{i,t} + \beta_4 Black_{i,t} + \beta_5 Hispanic_{i,t} + \beta_6 OtherRace_{i,t} + \varepsilon_{i,t}$

where $\varepsilon_{i,t} \sim N(0, \sigma^2)$.

The likelihood function for this linear regression model with normal errors is:

$$\mathcal{L}(\alpha, \beta, \sigma^2 | y, X) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - X_i\beta)^2}{2\sigma^2}\right)$$

The log-likelihood function is:

$$\ell(\alpha, \beta, \sigma^2 | y, X) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - X_i\beta)^2$$

In practice, we minimize the negative log-likelihood:

$$-\ell(\alpha, \beta, \sigma^2 | y, X) = \frac{n}{2}\log(2\pi) + \frac{n}{2}\log(\sigma^2) + \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - X_i\beta)^2$$

Data and Sample Selection

The sample was restricted to:

- Male heads of household (hsex = 1)
- Age between 25 and 60 years
- Hourly wage greater than $7

Hourly wage was calculated as:

$$\text{hr\_wage} = \frac{\text{hlabinc}}{\text{hannhrs}}$$

and the dependent variable is ln(hr_wage).

Interpretation of β_1 (Returns to Education)

The coefficient β_1 represents the percentage increase in wages associated with an additional year of education. The estimated values for β_1 across different years are:

Year	OLS β_1	MLE β_1
1971	0.0665	0.0665
1980	0.0660	0.0660
1990	0.0955	0.0955
2000	0.1103	0.1103

Table 1: Returns to Education (β_1) Over Time

Interpretation

- In **1971**, an additional year of education increased wages by approximately **6.65%**
- In **1980**, returns to education remained stable at about **6.60%**
- By **1990**, the returns increased significantly to approximately **9.55%**
- In **2000**, returns to education rose further to about **11.03%**

It indicates that returns to education were relatively flat during the 1970s-1980s but increased substantially during the 1990s and 2000s. This suggests that education became increasingly valuable in the labor market over this period.

Unit Test Verification

The unit test confirms that the Maximum Likelihood Estimation produces results identical to Ordinary Least Squares for this linear model, as expected theoretically. The test passes successfully, as shown below:

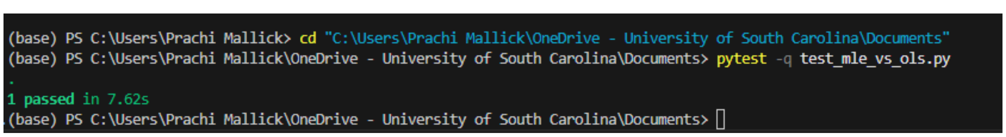

Figure 1: Unit Test Results Showing MLE and OLS Agreement